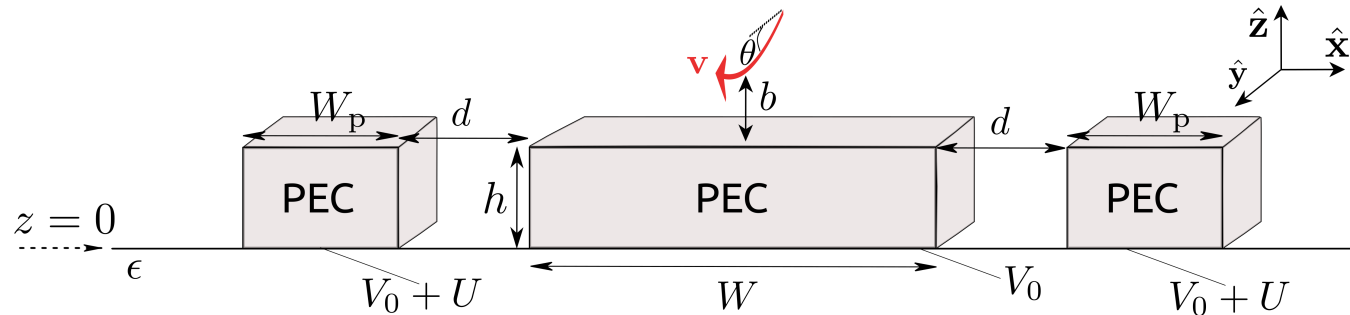


Electron-coupling to waveguide: virtual geometry

Add a substrate below the rectangular wg:

```
54 photon-energy E1eV E1eV NE
55 grid xmin xmax Nz zmin zmax Nz
56 incident-electron-z EeV 0 h+50
57
58 include rectangular_wg_substrate.bem
59
60 begin-virtual-structure
61 add-arc 33 11 0 0 R L+W/2 0 -180 0 N4
62
63 begin-calculation
64 calc EELS datfiles${NOFfiles}/${NOF}.dat
65 end-calculation
66 end
67
```

} virtual geometry

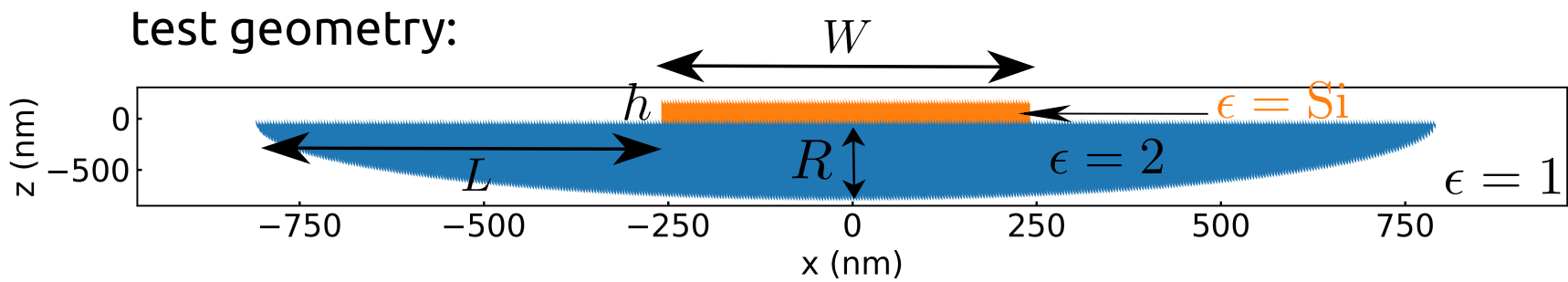


$R = W/2 + L$   $W = 500$

$h = 300$

$L = 500$

$E_e = 200 \text{ keV}$



the height of the peaks change with L but not the position of the modes

